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Cont2SAS: A python package for calculating SAS parameters from continuum nanostructures

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Summary

`Cont2SAS` is a software tool built around the existing software solution `Sassena` – known for calculating scattering patterns from simulated atomic structures. The goal of `Cont2SAS` is to provide a software similar to `Sassena`, but for calculating Small Angle Scattering data from simulated continuum nanostructures. `Cont2SAS` can calculate SAS patterns, i.e. SAS intensity (I) vs. scattering vector magnitude (Q) using a novel numerical method. `Cont2SAS` can also calculate effective cross-section (σ_{eff}), i.e. the count rate of scattered radiation per incident unit flux. σ_{eff} is calculated by integrating instrument-agnostic SAS patterns taking instrument geometry into account. `Cont2SAS` can be used for different purposes, such as validating simulations, tuning simulation parameters, and analyzing SAS data.

Statement of need

Small Angle Scattering (SAS) experiments with neutrons (SANS) and X-rays (SAXS) are useful techniques for probing material nanostructures (Chen et al., 2012). With the Time Resolved (TR) variant of SAS, the time evolution of nanostructures can also be probed (Hollamby, 2013). However, a direct retrieval of nanostructure from SAS data is not possible due to the so-called phase problem, i.e. the phase loss in measured intensities (Billinge & Levin, 2007). Therefore, it is fruitful to combine SAS data with simulations to study nanostructures (Majumdar et al., 2024). This approach is also useful for validating theories underlying simulations (Reich et al., 2022).

Simulated real-space structures can be generated using atomistic or continuum simulation. However, atomistic simulation cannot feasibly simulate large structures (e.g. micrometer scale) over long times (e.g. seconds) due to computational limitations (Ghavanloo et al., 2019). This becomes problematic when the nanostructure evolution is influenced by larger length-scale structures or when the SAS data is recorded over long time periods. In such cases, continuum simulation is the only option. Despite this indispensability, there is no software available for calculating SAS data from continuum simulations. `Cont2SAS` aims to fill this void by providing a tool that offers fast and error-free comparison of continuum simulations and SAS data. The ultimate goal is to study nanomaterials and validating theories using SAS data, particularly when length-scales and time-scales, inaccessible to atomistic simulations, need to be simulated.

State of the field

From atomistic simulations, different software such as `nMoldyn` (Hinsen, 2023; Róg et al., 2003), `MDANSE` (Goret et al., 2017; Neutron & Facility, 2023), `LiquidLib` (Walter et al., 2018; Z-laboratory, 2023), and `Sassena` (Lindner, 2012, 2017; Lindner & Smith, 2012; Majumdar et al., 2024; Majumdar & Lindner, 2023) can calculate SAS patterns but they cannot calculate the same from continuum simulations. An in-house code is available for continuum simulations but it was built for a 2D membrane system lacking generality (Dorrell et al., 2020). Hence, `Cont2SAS` is built using `Sassena` as a backend calculator. `Sassena` was chosen over other software due to its notable computation speed and robustness (Majumdar et al., 2024).

Software features

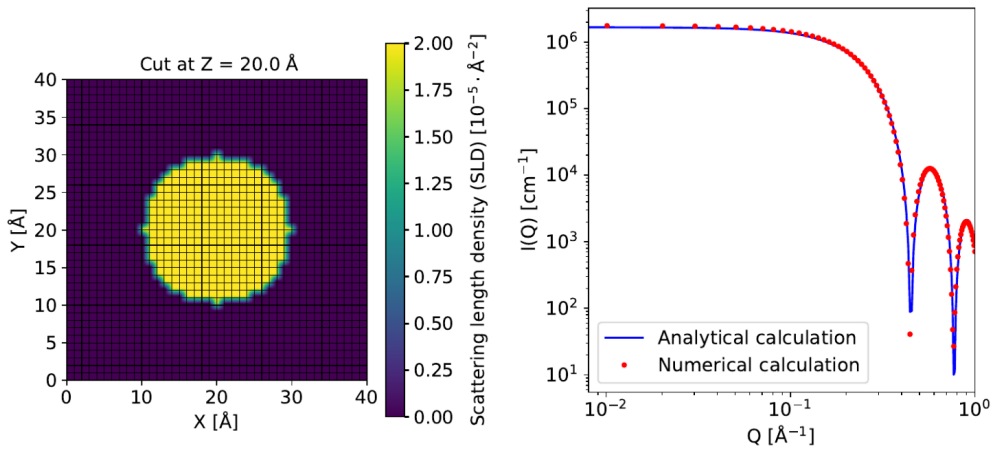

Figure 1: Workflow of SAS pattern calculation: [left] SLD assignment for a spherical nanoparticle on generated mesh and [right] numerical calculation of SAS pattern. The numerical calculation is validated against known analytical formula for spherical nanoparticles (Guinier et al., 1955).

`Cont2SAS` calculates SAS patterns taking simulated nanostructures as an input. The simulated structure must include a Scattering Length Density (SLD) distribution, which is either directly obtained from simulation or calculated from a set of simulated variables (e.g. molar density and molar fraction) before being provided as input to `Cont2SAS`. Simulating SLD distribution within `Cont2SAS` is also possible for model structures. The simulated input is processed to data tailor-made for `Sassena`. `Sassena` calculates SAS pattern, i.e. I vs Q data, as shown in Figure 1. The Q-dependence of I reveals the average shape and size of the nanoparticles, whereas the magnitude of I is related to the scattering contrast, i.e. the average diffrence in SLD between particles and their surrounding medium.

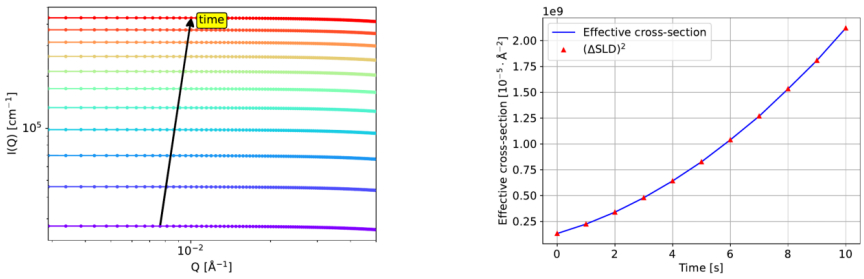

Figure 2: [left] TR-SAS data, i.e. SAS patterns at different timesteps ranging from 0 to 10 seconds. With time, the SLD of a spherical nanoparticle increases linearly. [right] σ_{eff} calculated from SAS patterns. The calculated σ_{eff} values are proportional to the $(\Delta\text{SLD})^2$, i.e. square of SLD difference between the particle and its environment.

For a simulation with more than one timestep, SAS patterns can be calculated from different time steps generating a TR-SAS data, as shown in Figure 2. From TR-SAS data, `Cont2SAS` can calculate the time evolution of effective cross-section (σ_{eff}), which is defined as the count rate per incident unit flux (see Figure 2). This feature was not available in the in-house code (Dorrell et al., 2020) or `Sassena` (Majumdar & Lindner, 2023). The time evolution of count rate is useful when the SLD of nanostructure changes over time without any change in average shape and size, e.g. while storing hydrogen in ball-milled powder sample (Aslan et al., 2019). The calculated σ_{eff} must be multiplied by an empirical factor before comparing with measured count rate.

Conclusion

`Cont2SAS` provides a software platform for calculating SAS pattern from continuum simulations of nanostructures. The addition of effective cross-section in the software package further enables the analysis of count rate. `Cont2SAS` can be used for analyzing SAS data and validating simulations to study nanomaterials.

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Declaration

During the preparation of this work, the authors used ChatGPT in order to improve the quality of code and article. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published code and article.

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